Practice Problems

Equations used:

Vibrational states in molecules:

$$E_N = (N + \frac{1}{2})\hbar\omega; \omega = \sqrt{\frac{k}{m}} = 2\pi f$$

Use harmonic mean for mass

$$m = \left(\sum \frac{1}{m_i}\right)^{-1} \left[= \frac{m_1 \cdot m_2}{m_1 + m_2}\right]$$

Rotational energy also exists, from Q.N. L

$$E_L = \frac{L(L+1)\hbar^2}{2mR_{eq}^2}$$

For dilute gas, go straight to Maxwell-Boltzmann.

$k = 1.38 \times 10^{-23}\,\mathrm{J/K} = 8.617 \times 10^{-5}\,\mathrm{eV/K}$

$$K_{avg} = \frac{3}{2}kT; N(E) = \frac{2N}{\sqrt{\pi}(kT)^{3/2}} E^{\frac{1}{2}} e^{-\frac{E}{kT}}$$

Approximate tiny distribution integrals.

$$\int_{E_m}^{E_m + \Delta E} N(E)dE \approx N(E) * \Delta E$$

Relativity Δt_0 is proper time; L_0 = proper length

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - u^2/c^2}}; L = L_0\sqrt{1 - u^2/c^2}$$

$$v = \frac{v' - u}{1 + v'u/c^2}; f' = f\sqrt{\frac{1 - u/c}{1 + u/c}}$$

Lorentz Transforms

$$x' = \frac{x - ut}{\sqrt{1 - u^2/c^2}}; \begin{matrix} y' = y; \\ z' = z; \end{matrix} t' = \frac{t - (u/c^2)x}{\sqrt{1 - u^2/c^2}}$$

$$v_x' = \frac{v_x - u}{1 - v_x u/c^2}; v_y' = \frac{v_y\sqrt{1 - u^2/c^2}}{1 - v_x u/c^2} v_z' = \frac{v_z\sqrt{1 - u^2/c^2}}{1 - v_x u/c^2}$$

Relativity alters momentum and energy

Problem 1

What atoms have the ground state configurations: (a) $1s^2 2s^2 2p^1$, (b) $1s^2 2s^2 2p^6 3s^1$, (c) $[Ne]3s^2 3p^5$

Solution

(a) Boron (B)

(b) Sodium (Na)

(c) Chlorine (Cl)

Problem 2

What is the ground state configuration of the ion K^+?

Solution

$\boxed{1s^2 2s^2 2p^6 3s^2 3p^6}$

Problem 3

What is the ground state configuration of the ion S^{-2}?

Solution

$\boxed{1s^2 2s^2 2p^6 3s^2 3p^6}$

Problem 4

What is the electron configuration of the first excited state of He?

Solution

$\boxed{1s^1 2s^1}$

Problem 5

For the hydrogen atom in the 3d excited state, give the possible values of (ℓ, m_ℓ, s, m_s) for the atom.

Solution

$\ell = 2$; $m_\ell \in \{-2, -1, 0, 1, 2\}$; $s = \frac{1}{2}$; $m_s \in \left\{\frac{1}{2}, -\frac{1}{2}\right\}$

Problem 6

List the quantum numbers (n, ℓ, m_ℓ, m_s) for all the electrons in a ground state nitrogen atom.

Solution

1s	$(1, 0, 0, -\frac{1}{2})$	$(1, 0, 0, \frac{1}{2})$	
2s	$(2, 0, 0, -\frac{1}{2})$	$(2, 0, 0, \frac{1}{2})$	
2p	$(2, 1, -1, \frac{1}{2})$	$(2, 1, 0, \frac{1}{2})$	$(2, 1, 1, \frac{1}{2})$

Problem 7

Infrared photons of wavelength 3440 nm are observed in the emission spectrum of molecular HCl. You suspect that this is from excitations of vibrational modes of the molecule as a harmonic oscillator. The mass of H is 1.008 u and the mass of Cl is 34.97 u where the atomic mass unit $u = 1.63 \times 10^{-27}$ kg. What is the 'spring constant' of this ionic bond?

Solution

Energy of a photon can be computed from wavelength, so let's do that.

$$E = \frac{hc}{\lambda} = \frac{1240\,\text{eV}\cdot\text{nm}}{3440\,\text{nm}} = 0.36\,\text{eV} \times \frac{1.602 \times 10^{-19}\,\text{J}}{1\,\text{eV}} = 5.7672 \times 10^{-20}\,\text{J}$$

This is the difference between two vibrational nodes' energies, or equivalent to $\hbar\omega$. Since $\omega = \sqrt{\frac{k}{m}}$, we can from here calculate the spring constant k. Make sure to use the harmonic mean for the mass.

$$\begin{aligned}
E &= \hbar\omega = \hbar\sqrt{\frac{k}{m}} \\
m &= \frac{m_1 * m_2}{m_1 + m_2} = \frac{34.97 * 1.008}{34.97 + 1.008} = 0.9798 \\
k &= \frac{E^2 m}{\hbar^2} = \frac{(5.7672 \times 10^{-20}\,\text{J})^2(0.9798 \times 1.63 \times 10^{-27}\,\text{kg})}{(1.054 \times 10^{-34}\,\text{J}\cdot\text{s})^2} = \boxed{478\,\text{N/m}}
\end{aligned}$$

Problem 8

Consider the diatomic molecule H_2. You can model it as two point masses (each of mass 1.08 u where $1u = 931.5\,\mathrm{MeV/c^2}$) with a separation of 0.074 nm. What is the wavelength of the photons emitted when this 'quantum rotor' transitions with the lowest possible energy change?

Solution

The miminum transition would come from transition between $L = 1$ to $L = 0$. We can calculate the energy transition here. Don't forget that we use the harmonic mean for the mass.

$$\begin{aligned}
E &= \frac{L(L+1)\hbar^2}{2mR_{eq}^2} = \frac{1(2)(hc)^2}{(2\pi)^2 2mR_{eq}^2 c^2} = \frac{2(1240\,\mathrm{eV\cdot nm})^2}{(2\pi)^2(931.5\,\mathrm{MeV})(0.074\,\mathrm{nm})^2} \\
&= \frac{2 * 1.5376\,\mathrm{eV}}{4\pi^2 931.5 * 5.476 \times 10^{-3}} = \frac{3.0752\,\mathrm{eV}}{64.1} = 0.048\,\mathrm{eV} \\
\lambda &= \frac{hc}{E} = \frac{1240\,\mathrm{eV\cdot nm}}{0.048\,\mathrm{eV}} = \boxed{81.2\,\mathrm{\mu m}}
\end{aligned}$$

Problem 9

Consider 1 mol of a dilute He gas at temperature 293 K. (a) What is the mean value of the energy, E_m, of an atom? (b) What fraction of the atoms has an energy in an interval of 0.01*E_m, centered at E_m?

Solution (a)

Use the Maxwell-Boltzmann energy distribution and find the average kinetic energy.

$$K_{avg} = \frac{3}{2}kT = \frac{3}{2} * 8.617 \times 10^{-5}\,\mathrm{eV/K} * 293\,\mathrm{K} = \boxed{37.87 \times 10^{-3}\,\mathrm{eV}}$$

Solution (b)

Recall the Maxwell-Boltzmann distribution.

$$N(E) = \frac{2N}{\sqrt{\pi}(kT)^{3/2}} E^{1/2} e^{E/kT}$$

Divide both sides by the number of total particles, which happens to already be in the equation.

$$\frac{N(E)}{N} = \frac{2}{\sqrt{\pi}(kT)^{3/2}} E^{1/2} e^{E/kT}$$

Multiply both by dE and take the integral over the span. Use the approximation.

$$\begin{aligned}
\int_{0.995\,E_m}^{1.005\,E_m} \frac{N(E)}{N}\,dE &\approx \frac{2}{\sqrt{\pi}(kT)^{3/2}} E_m^{1/2} e^{E_m/kT} * \Delta E \\
&= \frac{2}{\sqrt{\pi}(25.25 \times 10^{-3})^{3/2}} (37.87 \times 10^{-3})^{3/2} e^{-\frac{37.87\times 10^{-3}}{25.25\times 10^{-3}}} * 0.01 \\
&= \frac{32.89 \times 10^{-6}}{7.11 \times 10^{-3}} = 0.0046
\end{aligned}$$

This means about $\boxed{0.46\%}$ of all atoms are in that range.

Problem 10

The speed of light is the fastest speed that any signal can travel. We say that if two events (A and B) in space time can be connected by a signal that travels at or less than the speed of light then these two events can be causally connected. That is, the earlier event can cause the later event and all possible different observer frames will agree on which event was the earlier event. (Effects can not occur before causes)

(a) Show that if A and B are connected by a light signal in a space-time frame where A happens first, then no matter what other frame observes the events, B always happens later than A.

But if two events can only be connected by a signal that travels faster than light, which is impossible, then the two events cannot be causally connected- and there can be an observer frame that observes the events to happen simultaneously, a different observer frame observes event A to happen first and event B later; and another observer frame to observe that the event B happens first.

Pick two events A and B that would be connected by a signal that travels at twice the speed of light and pick A to have happened first and list them.

(b) What is the relative velocity of a frame where observers in that frame would see A and B happen at the same time? What are the relative velocities of frames that would see B happen first?

Solution (a)

If event A causes event B by a light signal, then A and B are colinear in a spacetime diagram along a line of slope 1. A would also take place before B.

$$t_B > t_A$$
$$0 < t_B - t_A = \Delta t_0$$

Suppose for contradiction that in some frame of reference, event B takes place at the same time or before event A.

$$t'_B \leq t'_A$$
$$t'_B - t'_A = \Delta t \leq 0$$

If we are in a relativistic frame of reference, we are traveling at a specific speed u. We can apply the Lorentz transformation for the time here.

$$\Delta t = \frac{\Delta t_0}{\sqrt{1 - u^2/c^2}} \leq 0$$
$$\Delta t_0 \leq 0$$
$$t_B - t_A \leq 0$$
$$t_B \leq t_A$$

This is a contradiction. As such, B always happens after A.

Solution (b)

Suppose that event A and B have (x, t) coordinates $(2\,\mathrm{c}\cdot\mathrm{yr}, 1\,\mathrm{yr})$ and $(4\,\mathrm{c}\cdot\mathrm{yr}, 2\,\mathrm{yr})$, respectively. We can apply the Lorentz transformation here.

$$t' = 0 = \frac{t - (u/c^2)x}{\sqrt{1 - u^2/c^2}}$$

For ths to be equal to zero, t must be equal to $(u/c^2)x$.

$$t = (u/c^2)x$$
$$u = \frac{tc^2}{x}$$

The value of $\frac{x}{t}$ is $2c$ in this instance. We can plug this in.

$$u = \frac{c^2}{2c} = \frac{c}{2}$$

If the frame is traveling at a speed of $\frac{c}{2}$, the events would appear to happen at the same time. Any faster and event A would appear to happen after event B.

Problem 11

Consider two protons headed towards each other with the same speeds. Each proton has mass $m_p = 938.8\,\mathrm{MeV/c^2}$. They collide and create a deuteron with mass $m_d = 1875.6\,\mathrm{MeV/c^2}$ and a positively charged particle called the 'pion plus'. The pion particle symbol is π^+ and its mass is $m_\pi = 139.6\,\mathrm{MeV/c^2}$. This interaction can be symbolized with $p^+ + p^+ \to \pi^+ + d^+$.

(a) What is the minimum kinetic energy that each proton must have in order for this to be able to happen?

(b) Say that the two protons head into the collision from opposite directions each with 100 MeV of kinetic energy. How much momentum will the pion have after it is created?

Solution (a)

For the minimum necessary velocity for genesis, we assume that upon being created, the two new particles will be completely immobile, so no kinetic energy. This means the only energy after creation is their rest energy, which we can calculate.

$$E_{net} = m_\pi c^2 + m_d c^2 = 139.6\,\mathrm{MeV} + 1875.6\,\mathrm{MeV} = 2015.2\,\mathrm{MeV}$$

From here, divide it by two and subtract the rest energy of the proton to find the necessary kinetic energy of each photon.

$$\begin{aligned} E_{p^+} &= \frac{2015.2}{2}\,\mathrm{MeV} = 1007.6\,\mathrm{MeV} \\ K = E_{p^+} - E_0 &= 1007.6\,\mathrm{MeV} - 938.8\,\mathrm{MeV} = \boxed{68.8\,\mathrm{MeV}} \end{aligned}$$

Solution (b)

The deuterion, if created immobile, will have only its rest energy, which is the rest energy of the two protons that created it. This means the 200 MeV of energy (100 eV from each proton) will all go to the pion.

$$\begin{aligned} E^2 &= (pc)^2 + (mc^2)^2 \\ p^2 &= \left(\frac{E}{c}\right)^2 - (mc)^2 = (200\,\mathrm{MeV/c})^2 - (139.6\,\mathrm{MeV/c})^2 \\ &= 40 \times 10^{15}\,\mathrm{eV^2/c^2} - 19.488 \times 10^{15}\,\mathrm{eV^2/c^2} = 20.512 \times 10^{15}\,\mathrm{eV^2/c^2} \\ p &= \sqrt{20.512 \times 10^{15}\,\mathrm{eV^2/c^2}} = \boxed{143\,\mathrm{MeV/c}} \end{aligned}$$

Chapter 8

Many Electron Atoms

Equations used:

$$Z_{eff} = q_{\text{nucleus}} - q_{\text{screening electrons}} \tag{8.1}$$

$$E_n = (-13.6\,\text{eV})\frac{Z_{eff}^2}{n^2} \tag{8.2}$$

For transition energies:

$$\Delta E = 13.6\,\text{eV}(Z-1)^2\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right) \tag{8.3}$$

For electrons in incomplete shells, exists angular (L) & magnetic (M_L) Q.N's:

$$|L_i| = \hbar\sqrt{l_i(l_i+1)} \tag{8.4}$$

$$L_{max} = l_1 + l_2; L_{min} = |l_1 - l_2| \tag{8.5}$$

$$L \in [L_{min}, L_{max}] \cap \mathbb{N} \tag{8.6}$$

$$M_L = \sum m_{l,i}; M_L \in [-L, L] \cap \mathbb{N} \tag{8.7}$$

S can be computed same as M_L but with spin quantum number
Hund's rule: $L = M_{L,max}$, $S = M_{S,max}$; maximize total spin
$\Delta L \in \{0, \pm 1\}; \Delta S = 0$
HeNe laser produces photons

$$\lambda = 632.8\,\text{nm}; E = 1.96\,\text{eV} \tag{8.8}$$

$$E_0 = \sqrt{\frac{2I}{c\varepsilon_0}}; I = \frac{P}{A} \tag{8.9}$$

8.1 Problem 1

(a) List the six possible sets of quantum numbers (n, l, m_l, m_s) of a 2p electron. (b) Suppose we have an atom such as carbon, which has two 2p electrons. Ignoring the Pauli principle, how many different possible combinations of quantum numbers of the two electrons are there? (c) How many of the possible combinations of part (b) are eliminated by applying the Pauli principle? (d) Suppose carbon is in an excited state with configuration $2p^1 3p^1$. Does the Pauli principle restrict the choice of quantum numbers for the electrons? How many different sets of quantum numbers are possible for the two electrons?

8.1.1 Solution (a)

$$\begin{array}{ccc} (2,1,-1,-\frac{1}{2}) & (2,1,0,-\frac{1}{2}) & (2,1,1,-\frac{1}{2}) \\ (2,1,-1,\frac{1}{2}) & (2,1,0,\frac{1}{2}) & (2,1,1,\frac{1}{2}) \end{array}$$

8.1.2 Solution (b)

Take the triangle number of six (for the six configurations).

$$T_6 = \frac{n(n+1)}{2} = \frac{6*7}{2} = \boxed{21} \tag{8.10}$$

8.1.3 Solution (c)

Only six, since there are six states and Pauli only excludes them when the two states are identical. This totals us at 15 states.

8.1.4 Solution (d)

The Pauli principle still applies, but it would nt be important here. The two have different values for quantum number n. Furthermore, we would no longer have to account for duplicates, so we can just multiply the number of possible combinations in each shell. This gives us $6 * 6 = \boxed{36}$ possible combinations.

8.2 Problem 3

(a) How many different sets of quantum numbers (n, l, m_l, m_s) are possible for an electron in the 4f level?

(b) Suppose a certain atom has three electrons in the 4f level. What is the maximum possible value of the total m_s of the three electrons?

(c) What is the maximum possible total m_l of three 4f electrons?

(d) Suppose an atom has ten electrons in the 4f level. What is the maximum possible value of the total m_s of the ten 4f electrons?

(e) What is the maximum possible total m_l of ten 4f electrons?

8.2.1 Solution (a)

$$2(2\ell + 1) = 2(2 * 3 + 1) = 2 * 7 = \boxed{14} \tag{8.11}$$

8.2.2 Solution (b)

Each of them could have a value of $m_s = \frac{1}{2}$. Multiplying that by three, we get $\underline{\frac{3}{2} = 1.5}$

8.2.3 Solution (c)

They would have m_l values of 3, 3, and 2 first. The total would be $\boxed{8}$.

8.2.4 Solution (d)

Start by filling it with seven electrons where $m_s = \frac{1}{2}$. This leaves us with a total m_s of $\frac{7}{2}$ and three electrons remaining. All three of these remaining electrons would have to have $m_s = -\frac{1}{2}$. Adding the two together, the total value would be $\boxed{2}$.

8.2.5 Solution (e)

Two electrons would have values of 3, two of 2, two of 1, etc. all the way down to and including -1.

$$2\,(3 + 2 + 1 + 0 - 1) = 2 * 5 = \boxed{10} \tag{8.12}$$

8.3 Problem 5

(a) List all elements with a p^3 configuration.

(b) List all elements with a d^7 configuration.

8.3.1 Solution (a)

Nitrogen, Phosphorus, Arsenic, Antimony, and Bismuth, plus whatever has atomic number 115.

8.3.2 Solution (b)

Cobalt, Rhodium, Iridium, and Meitnerium.

8.4 Problem 7

(a) What is the electronic configuration of Fe?

(b) In its ground state, what is the maximum possible total m_s of its electrons?

(c) When the electrons have their maximum possible total m_s, what is the maximum total m_l?

(d) Suppose one of the d electrons is excited to the next highest level. What is the maximum possible total m_s, and when m_s has its maximum total what is the maximum total m_l?

8.4.1 Solution (a)

$\boxed{1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6}$

8.4.2 Solution (b)

Anything but the outermost shell ($3d$) would be empty and cancel out. At 6 electrons, we would have at most five electrons with $m_s = \frac{1}{2}$ and at least one electron with $m_s = -\frac{1}{2}$. This has a total m_s of $\boxed{2}$.

8.4.3 Solution (c)

Subshells with $m_l \in \{+2\}$ would be full (two electrons). Subshells with $m_l \in \{+1, 0, -1, -2\}$ would have only one electron.

$$2 * 2 + (1 + 0 - 1 - 2) = 4 - 2 = \boxed{2} \tag{8.13}$$

8.4.4 Solution (d)

To achieve the maximum value of m_s, we would have to take our answer from (b) move an electron from $m_s = -\frac{1}{2}$ to $m_s = \frac{1}{2}$. This would have the total value of $m_s = \boxed{3}$. The next energy level that would be accessible by an electron would be 4p, so the maximum possible value of m_l would be 1. Add everything together and make the replacement.

$$2 + 1 + 1 + 0 - 1 - 2 = \boxed{1} \tag{8.14}$$

8.5 Problem 9

The ground state of neutral beryllium (Z = 4) is $1s^2 2s^2$. Use the electron screening model to predict the energies of the following excited states: (a) $1s^2 2s^1 3p^1$ (measured $-2.02\,\mathrm{eV}$) and (b) $1s^2 2s^1 4d^1$ ($-0.90\,\mathrm{eV}$).

8.5.1 Solution (a)

Here, we use equation 8.1.

$$E_n = (-13.6\,\mathrm{eV})\frac{Z_{eff}^2}{n^2} = (-13.6\,\mathrm{eV})\frac{(4-3)^2}{3^2} = (-13.6\,\mathrm{eV})\frac{1}{9} = \boxed{-1.5\,\mathrm{eV}} \tag{8.15}$$

8.5.2 Solution (b)

$$E_n = (-13.6\,\mathrm{eV})\frac{Z_{eff}^2}{n^2} = (-13.6\,\mathrm{eV})\frac{(4-3)^2}{4^2} = (-13.6\,\mathrm{eV})\frac{1}{16} = \boxed{-0.85\,\mathrm{eV}} \tag{8.16}$$

8.6 Problem 11

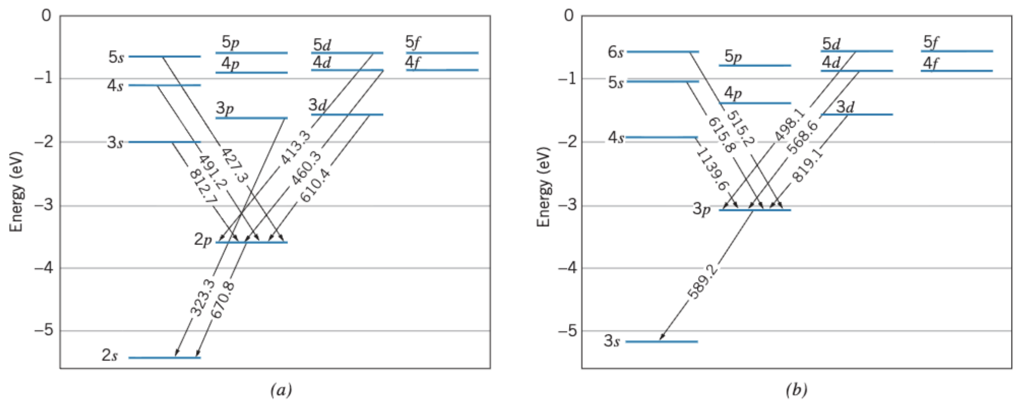

Figure 8.4

(a) Using the information for lithium given in Figure 8.4, compute the energy difference of the 3p and 3d states.

(b) Compute the energy of the 3s, 4s, and 5s states above the ground state.

(c) The ionization energy of lithium in its ground state is 5.39 eV. What is the ionization energy of the 2p state? Of the 3s state?

8.6.1 Solution (a)

3p to 2s emits a photon of wavelength $\lambda = 323.3\,\text{nm}$. 3d to 2p emits a photon of wavelength $\lambda = 610.4\,\text{nm}$. 2p to 2s emits a photon of wavelength $\lambda = 670.8\,\text{nm}$. From here, make a pathway from 3d to 3p, ensuring to make everything in terms of energy.

$$E_{3d\to 3p} = E_{3d\to 2p} + E_{2p\to 2s} - E_{3p\to 2s} = hc\left(\frac{1}{\lambda_{3d\to 2p}} + \frac{1}{\lambda_{2p\to 2s}} - \frac{1}{\lambda_{3p\to 2s}}\right) \tag{8.17}$$

$$= (1240\,\text{eV}\cdot\text{nm})\left(\frac{1}{610.4\,\text{nm}} + \frac{1}{670.8\,\text{nm}} - \frac{1}{323.3\,\text{nm}}\right) \tag{8.18}$$

$$= (1240\,\text{eV})\left(1.64\times 10^{-3} + 1.49\times 10^{-3} - 3.09\times 10^{-3}\right) \tag{8.19}$$

$$= 0.04 * 1240\,\text{eV} = \boxed{49.6\,\text{eV}} \tag{8.20}$$

8.6.2 Solution (b)

Assume

8.7 Problem 13

Compute the K_α X ray energies of calcium (Z = 20), zirconium (Z = 40), and mercury (Z = 80). Compare with the measured values of 3.69 keV, 15.8 keV, and 70.8 keV. (See Question 16).

8.7.1 Solution

Calcium: K_α is the transition between $n = 2$ and $n = 1$.

$$\Delta E = (13.6\,\mathrm{eV})(Z-1)^2\left(\frac{1}{n_i^2}-\frac{1}{n_f^2}\right) = (13.6\,\mathrm{eV})19^2\left(\frac{1}{2^2}-\frac{1}{1^2}\right) = \boxed{3.6822\,\mathrm{keV}} \tag{8.21}$$

This is not far off from the measured value.

Zirconium:

$$\Delta E = (13.6\,\mathrm{eV})(Z-1)^2\left(\frac{1}{n_i^2}-\frac{1}{n_f^2}\right) = (13.6\,\mathrm{eV})39^2\left(\frac{1}{2^2}-\frac{1}{1^2}\right) = \boxed{15.5142\,\mathrm{keV}} \tag{8.22}$$

This is not far off from the measured value.

Mercury:

$$\Delta E = (13.6\,\mathrm{eV})(Z-1)^2\left(\frac{1}{n_i^2}-\frac{1}{n_f^2}\right) = (13.6\,\mathrm{eV})79^2\left(\frac{1}{2^2}-\frac{1}{1^2}\right) = \boxed{63.6582\,\mathrm{keV}} \tag{8.23}$$

This is much farther off from the measured value.

8.8 Problem 15

Use Hund's rules to find the ground-state L and S of (a) Ce, configuration [Xe]$6s^2 4f^1 5d^1$; (b) Gd, configuration [Xe]$6s^2 4f^7 5d^1$; (c) Pt, configuration [Xe]$6s^1 4f^{14} 5d^9$.

8.8.1 Solution (a)

Hund's rule says that $L = M_{L,max}$ and $S = M_{S,max}$.

The electrons in the outermost shells are $4f^1 5d^1$. The maximum posible values for m_l are 3 for the f-orbital and 2 for the d-orbital. Together, they total $L = M_{L,max} = 3 + 2 = \boxed{5}$.

The electrons in the outermost shells are $4f^1 5d^1$. The maximum posible values for m_s are $\frac{1}{2}$ both. Together, they total $S = M_{S,max} = \frac{1}{2} + \frac{1}{2} = \boxed{1}$.

8.8.2 Solution (b)

The electrons in the outermost shells are $4f^7 5d^1$. The maximum posible values for m_l are two copies of 3, 2, and 1 and one copy of 0 from the $4f$ orbital, and one copy of 2 for the d-orbital. Together, they total $L = M_{L,max} = 2(3 + 2 + 1) + 2 + 0 = \boxed{14}$.

The electrons in the outermost shells are $4f^1 5d^1$. At most, all 7 electrons in the $4f$ orbital would have value $\frac{1}{2}$, as with the one electron in the $5d$ orbital. Together, they total $S = M_{S,max} = 8 * \frac{1}{2} = \boxed{4}$.

8.8.3 Solution (c)

Here, the $4f$-orbital is full, so we can disregard that. In the $6s$ orbital, the maximum possible value of m_l is just 0. In the $5d$ orbital, the maximum possible value of M_L would come when only one value of m_l is not taken up, being -2, which would make the total sum $L = \boxed{2}$.

For a similar reason, the maximum possible value of M_S is $S = \boxed{1}$.

8.9 Problem 17

A certain excited state of an atom has the configuration $4d^1 5d^1$. What are the possible L and S values?

8.10 Problem 19

A small helium-neon laser produces a light beam with an average power of 3.5 mW and a diameter of 2.4 mm. (a) How many photons per second are emitted by the laser? (b) What is the amplitude of the electric field of the light wave? Compare this result with the electric field at a distance of 1 m from an incandescent light bulb that emits 100 W of visible light.

8.10.1 Solution (a)

A watt is equivalent to a joule per second. Since this is one second only, we can just multiply the power by one second to get the energy from one second to be $E = 3.5\,\mathrm{mJ}$. Each HeNe laser photon has an energy of $1.96\,\mathrm{eV}$, or 3.13×10^{-19} J. Dividing the energy delivered by the energy of each photon, we get $\boxed{1.11 \times 10^{16}}$ photons.

8.10.2 Solution (b)

First find the intensity.

$$I = \frac{P}{A} = \frac{3.5\,\mathrm{mW}}{\pi(1.2\,\mathrm{mm})^2} = 773.6\,\mathrm{W/m^2} \tag{8.24}$$

$$E_0 = \sqrt{\frac{2I}{c\varepsilon_0}} = \sqrt{\frac{2 * 773.6\,\mathrm{W/m^2}}{2.998 \times 10^8\,\mathrm{m/s} * 8.85 \times 10^{-12}\,\mathrm{C^2/(Nm^2)}}} \tag{8.25}$$

$$= \sqrt{\frac{1547.2}{2.65 \times 10^{-3}}}\,\mathrm{N/C} = \sqrt{583900}\,\mathrm{N/C} = \boxed{764\,\mathrm{N/C}} \tag{8.26}$$

Chapter 9

Molecular Structure

Equations used:

$$E(\mathrm{H_2}^+) = -16.3\,\mathrm{eV}; E(\mathrm{H_2}) = -31.7\,\mathrm{eV}$$
$$N_A = 6.022 \times 10^{23}\,\mathrm{mol}^{-1}$$
$$1\,\mathrm{eV} = 1.602 \times 10^{-19}\,\mathrm{J}$$
$$1\,\mathrm{J} = 6.242 \times 10^{18}\,\mathrm{eV}$$
$$U_C = -\frac{q_1 q_2}{(4\pi\varepsilon_0)R_{eq}}$$
$$\hbar c = 197.3\,\mathrm{eV}\cdot\mathrm{nm}; hc = 1240\,\mathrm{eV}\cdot\mathrm{nm}$$
$$c^2 = 931.5\,\mathrm{MeV/amu}$$
$$|\Delta N| = |\Delta L| = 1$$

9.1 Problem 1

Calculate the ionization energy of H_2.

9.1.1 Solution

The ionization energy of Hydrogen is the energy of ${H_2}^+$ minus the energy of H_2.

$$E({H_2}^+) + E(H_2) = -16.3\,\text{eV} + 31.7\,\text{eV} = \boxed{15.4\,\text{eV}} \tag{9.1}$$

9.2 Problem 3

Bond strengths are frequently given in units of kilojoules per mole. Find the molecular dissociation energy (in eV) from the following bond strengths: (a) NaCl, 410 kJ/mole; (b) Li_2, 106 kJ/mole; (c) N_2, 945 kJ/mole.

9.2.1 Solution (a)

Just convert the energy to that for an individual bond.

$$E = 410\,\text{kJ/mol} \times \frac{1000\,\text{kJ}}{1\,\text{J}} \times \frac{1\,\text{mol}}{6.022 \times 10^{23}} \times 6.242 \times 10^{18}\,\text{eV/J} \tag{9.2}$$

$$= 6.81 \times 10^{-19}\,\text{J} \times 6.242 \times 10^{18}\,\text{eV/J} = \boxed{4.25\,\text{eV}} \tag{9.3}$$

9.2.2 Solution (b)

$$E = 106\,\text{kJ/mol} \times \frac{1000\,\text{kJ}}{1\,\text{J}} \times \frac{1\,\text{mol}}{6.022 \times 10^{23}} \times 6.242 \times 10^{18}\,\text{eV/J} \tag{9.4}$$

$$= 1.76 \times 10^{-19}\,\text{J} \times 6.242 \times 10^{18}\,\text{eV/J} = \boxed{1.10\,\text{eV}} \tag{9.5}$$

9.2.3 Solution (c)

$$E = 945\,\text{kJ/mol} \times \frac{1000\,\text{kJ}}{1\,\text{J}} \times \frac{1\,\text{mol}}{6.022 \times 10^{23}} \times 6.242 \times 10^{18}\,\text{eV/J} \tag{9.6}$$

$$= 1.569 \times 10^{-18}\,\text{J} \times 6.242 \times 10^{18}\,\text{eV/J} = \boxed{9.80\,\text{eV}} \tag{9.7}$$

9.3 Problem 5

For each of the following pairs of molecules, predict which will have the larger dissociation energy: (a) Li_2 and Be_2; (b) B_2 and C_2; (c) CO and O_2. Check your prediction against tabulated values of the bond strengths or dissociation energies.

9.4 Problem 7

Calculate the Coulomb energy of KBr at the equilibrium separation distance.

9.4.1 Solution

Looking at Table 9.5, we know that the equilibrium separation of KBr is 0.282 nm. The Potassium and the bromine would both have an ionized charge of ± 1. We have an equation for the coulombic energy from the equilibrium separation.

$$U_C = -\frac{q_1 q_2}{(4\pi\varepsilon_0)R_{eq}} = \frac{e^2}{(4\pi\varepsilon_0)R_{eq}} = \frac{(1.602\times 10^{-19})^2}{(4\pi * 8.854\times 10^{-12}) * 0.282\,\mathrm{nm}} \tag{9.8}$$

$$= \boxed{8.18\times 10^{-19}\,\mathrm{J}} = \boxed{5.11\,\mathrm{eV}} \tag{9.9}$$

9.5 Problem 9

(a) Assuming a separation of 0.193 nm, compute the expected electric dipole moment of NaF. (b) The measured dipole moment is $27.2\times 10^{-30}\,\mathrm{C}\cdot\mathrm{m}$. What is the fractional ionic character of NaF?

9.6 Problem 11

The equilibrium separation of BaO is 0.194 nm and the measured electric dipole moment is $26.5\times 10^{-30}\,\mathrm{C}\cdot\mathrm{m}$. Calculate the fractional ionic character, assuming two valence electrons.

9.7 Problem 13

The vibrational transition in BeO is observed at a wavelength of 6.724 µm. What is the effective force constant of BeO?

9.7.1 Solution

We can convert wavelength to energy.

$$E = \frac{hc}{\lambda} \tag{9.10}$$

That can be related to the angular frequency, which can be related to the effective force constant.

$$\frac{hc}{\lambda} = E = \hbar\omega \tag{9.11}$$

$$\omega = \sqrt{\frac{k}{m}} = \frac{hc}{\hbar\lambda} = \frac{2\pi c}{\lambda} \tag{9.12}$$

$$k = \frac{4\pi^2 mc^2}{\lambda^2} \tag{9.13}$$

Take the harmonic mean of the masses to find the mass used here. From there, we can

$$m = \frac{m_1 m_2}{m_1 + m_2} = \frac{9.012 \times 16.00}{9.012 + 16.00} \times \frac{1\,\text{amu}}{6.022 \times 10^{26}\,\text{kg}} \tag{9.14}$$

$$= \frac{144.192}{25.012} \times \frac{1\,\text{amu}}{6.022 \times 10^{26}\,\text{kg}} = 5.765 \times \frac{1\,\text{amu}}{6.022 \times 10^{26}\,\text{kg}} = 9.573 \times 10^{-27}\,\text{kg} \tag{9.15}$$

$$k = \frac{4\pi^2 * 9.573 \times 10^{-27}\,\text{kg} * (2.998 \times 10^8)^2}{(6.724\,\mu\text{m})^2} = \boxed{751\,\text{N/m}} \tag{9.16}$$

9.8 Problem 15

The vibrational energy of CO is 0.2691 eV when the constituents of the molecule are the most abundant isotopes of carbon (m = 12.00 u) and oxygen (m = 16.00 u). (a) What would be the vibrational energy if the oxygen were replaced by the less abundant isotope with m = 18.00 u? (b) What would be the vibrational energy if the carbon in the original CO were replaced with radioactive carbon (used in radiocarbon dating) with mass 14.00 u?

9.9 Problem 17

Following the method of Example 9.4, calculate the energies and wavelengths of the three lowest rotational transitions emitted by molecular NaCl. In what region of the electromagnetic spectrum would these radiations be found?

9.9.1 Solution

Example 9.4 starts with finding the rotational quantity B. For NaCl, the effective mass can be found using the harmonic mean. Its equivalent radius, given in table 9.5, is 0.236 nm.

$$m = \frac{m_1 m_2}{m_1 + m_2} = \frac{22.99 * 35.45}{22.99 + 35.45}\,\text{amu} = 13.95\,\text{amu} \tag{9.17}$$

$$B = \frac{\hbar^2}{2mR_{eq}^2} = \frac{\hbar^2 c^2}{2mc^2 R_{eq}^2} = \frac{(197.3\,\text{eV}\cdot\text{nm})^2}{2 * 13.95\,\text{amu} * 931.5\,\text{MeV/amu} * (0.236\,\text{nm})^2} \tag{9.18}$$

$$= \frac{38927.29\,\text{eV}}{1.447\times 10^9} = 26.89\ \times 10^{-6}\,\text{eV} \tag{9.20}$$

We can find the wavelength through the energy, and we can find the energy through the rotational quantity and even numbers.

$$\lambda_n = \frac{hc}{E_n} = \frac{hc}{2nB} = \frac{1}{n}\cdot\frac{1240\,\text{eV}\cdot\text{nm}}{2 * 26.89\ \times 10^{-6}\,\text{eV}} = \frac{23.06}{n}\,\text{mm} \tag{9.21}$$

$$\lambda_1 = \frac{hc}{E_1} = \frac{23.06}{1}\,\text{mm} = \boxed{23.06\,\text{mm}} \tag{9.24}$$

$$\lambda_2 = \frac{hc}{E_2} = \frac{hc}{4B} = \frac{23.06}{2}\,\text{mm} = \boxed{11.53\,\text{mm}} \tag{9.25}$$

$$\lambda_3 = \frac{hc}{E_3} = \frac{hc}{6B} = \frac{23.06}{3}\,\text{mm} = \boxed{7.686\,\text{mm}} \tag{9.26}$$

These are radiowaves or radar.

9.10 Problem 19

Find the difference in the rotational parameter B of HCl for the two different masses of Cl (35 u and 37 u).

9.11 Problem 21

Figure 9.30 illustrates the combined rotational-vibrational structure when the vibrational energy is much larger than the rotational energy. Make a sketch showing the reverse situation, in which the rotational energy is much larger than the vibrational energy. Use a scale in which B = 10 units and hf = 2 units; show rotational levels up to L = 3 and vibrational levels to N = 3.

9.12 Problem 23

Based on the energy levels given by Eq. 9.19, calculate the energies of the emitted photons.

$$E_{NL} = (N + \frac{1}{2})hf + BL(L+1) \tag{9.19}$$

9.12.1 Solution

We have stats known about the value of the change in the constants N and L.

$$|\Delta N| = |\Delta L| = 1 \rightarrow \Delta N = \Delta L = \pm 1 \tag{9.27}$$

Suppose that $\Delta L = L_f - L$ and $\Delta N = N_f - N$. This gives us values for L_f and N_f relative to L and N.

$$L_f = L \pm 1; N_f = N \pm 1 \tag{9.28}$$

We can plug these in.

$$\begin{aligned}
\Delta E = E_f - E_i &= (N_f + \frac{1}{2})hf + BL_f(L_f+1) - (N + \frac{1}{2})hf - BL(L+1) && (9.29)\\
&= (N \pm 1 + \frac{1}{2})hf + B(L \pm 1)(L \pm 1 + 1) - (N + \frac{1}{2})hf - BL(L+1) && (9.30)\\
&= \pm hf + B(L(L \pm 1 + 1) \pm (L \pm 1 + 1)) - BL(L+1) && (9.31)\\
&= \pm hf + BL(L+1) \pm BL \pm B(L \pm 1) \mp B - BL(L+1) && (9.32)\\
&= \pm hf \pm BL \pm B(L \pm 1) \mp B = \pm hf \pm BL \pm BL \mp B \mp B && (9.33)\\
&= \boxed{\pm hf \pm 2B(L-1)} && (9.34)
\end{aligned}$$

9.13 Problem 25

Derive Eq. 9.23 from Eq. 9.22.

$$p(E_{NL}) = (2L+1)e^{-\frac{E_{NL}}{kT}} = (2L+1)e^{-\frac{(N+\frac{1}{2})hf+BL(L+1)}{kT}} \tag{9.22}$$

$$2L+1 = \sqrt{\frac{2kT}{B}} \tag{9.23}$$

9.14 Problem 27

The most intense absorption line in the rotational-vibrational spectrum of CO at room temperature occurs for L = 7. Justify this value with a calculation. (The equilibrium separation of CO is 0.113 nm.)

Chapter 10

Statistical Physics

Equations used:

$$TBD$$

10.1 Problem 1

A collection of three noninteracting particles shares 3 units of energy. Each particle is restricted to having an integral number of units of energy. (a) How many macrostates are there? (b) How many microstates are there in each of the macrostates? (c) What is the probability of finding one of the particles with 2 units of energy? With 0 units of energy?

10.2 Problem 3

Consider a system consisting of two particles, one with spin s = 1 and another with spin s = 1/2. (a) Considering a microstate to be an assignment of the z component of the spins of each of the particles, what is the total number of microstates of the two-particle system? (b) How many macrostates are there for the total spin of the two-particle system? (c) Find the number of microstates for each macrostate, and be sure the total number agrees with your answer to part (a).

10.2.1 Solution (a)

10.3 Problem 5

Calculate the probabilities given in Table 10.3 for E = 0 and E = 3 for (a) integral spin and (b) spin 1/2.